Operations Research 1: Deterministic Models
IP Modeling

Dr. Lei Zhao

Fall, 2025

Operations & Services Research (TOpS) Laboratory
Department of Industrial Engineering
Tsinghua University, China

Tatham Example

Tatham Example

- Seven (7) investments
  - Cash required and NPV
  - Total budget: $15,000

Table 6.1 Data for the Capital Budgeting Example

Investment	Cash Required	NPV
1	$5,000	$16,000
2	$2,500	$8,000
3	$3,500	$10,000
4	$6,000	$19,500
5	$7,000	$22,000
6	$4,500	$12,000
7	$3,000	$7,500

- Objective: maximize total NPV

Tatham Example

- Data

$$
\begin{aligned}
c_i &= \text{cash required of investment } i \\
N_i &= \text{NPV of investment } i \\
B &= \text{budget}
\end{aligned}
$$

- Decisions

$$
x_i = \begin{cases} 1, & \text{if we invest on investment } i, \\ 0, & \text{otherwise.} \end{cases}
$$

- Model

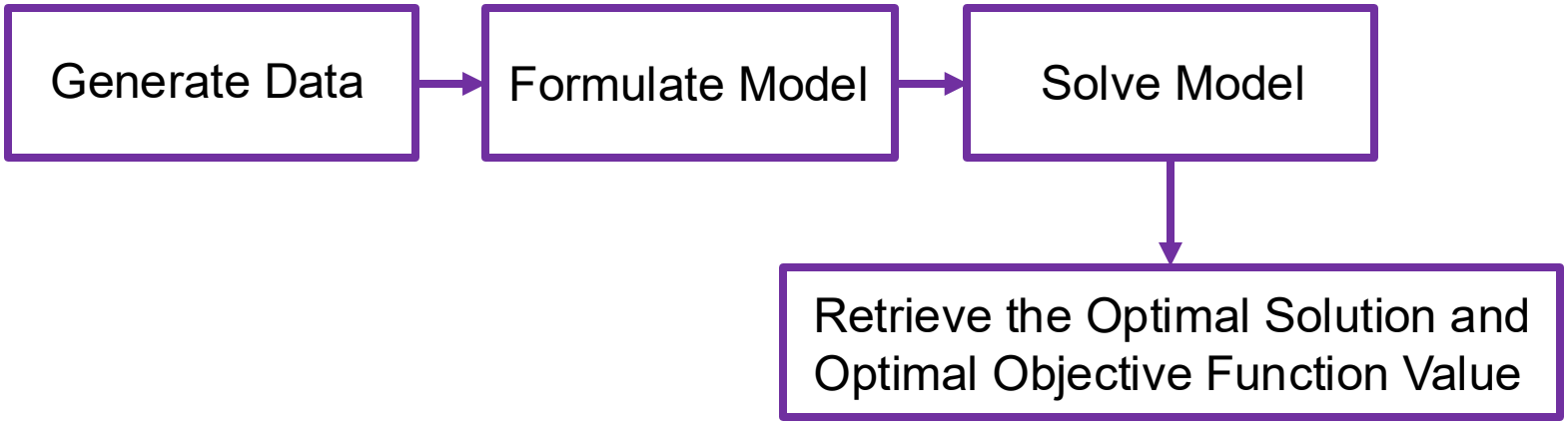
$$
\begin{array}{lll}
\max & \sum_{i \in I} N_i x_i & \text{maximize the total NPV} \\
\text{s.t.} & \sum_{i \in I} c_i x_i \leq B & \text{budget constraint} \\
 & x_i \text{ binary}, \forall i \in I & \text{equivalently, } x_i \in \{0, 1\}, \forall i \in I
\end{array}
$$

An Exemplary Program

- Notes
  - This exemplary program focuses on illustrating how to formulate and optimize an IP model using Gurobi in Python.
  - The main structure of the code is similar to that in the tutorial *LP Modeling*.
  - Pay attention to changes in the variable types.

An Exemplary Program

- main.py
  - First, read the Dictionary objects that contain the input data.
  - Next, formulate a Gurobi model.
    - Using the Dictionary objects.
  - Finally, solve the Gurobi model, retrieve the optimal solution and optimal objective function value.



An Exemplary Program

- model.py
  - Formulate and solve the Tatham example using Gurobi.
  - Functions
    - formulateModel(): Formulate a Gurobi model based on given Dictionary objects.
    - solveModel(): Optimize the given Gurobi model.
    - getOptimalSolution(): Retrieve the optimal solution and optimal objective function value.
    - writeModel(): Save the given Gurobi model to a file, either in MPS or LP format.
    - readModel(): Load a Gurobi model from an MPS or LP file.

Variable Types in Gurobi 12.0

- Available variable types when creating a variable using the function Model.addVar() or Model.addVars()
  - Continuous (GRB.CONTINUOUS)
  - General integer (GRB.INTEGER)
  - Binary (GRB.BINARY)
  - Semi-Continuous (GRB.SEMICONT)
  - Semi-Integer (GRB.SEMIINT)
  - Details:
    - https://docs.gurobi.com/projects/optimizer/en/current/reference/python/model.html#Model.addVar
    - https://docs.gurobi.com/projects/optimizer/en/current/reference/attributes/variable.html#vtype